

CIS 5500 — Data-Driven Shopping Assistant

Milestone 4: API Specification

Chenguang Shen Luis Garcia Leo Tan Anika Madan

April 20, 2026

1 Overview

This document specifies the backend API routes for the Data-Driven Shopping Assistant. Each route either maps to one of the 10 required SQL queries (plus two additional complex queries for the Value Rankings and Top Value views) or serves as an auxiliary helper for client-side UI (e.g., populating dropdowns). All routes are served by an Express.js application and return JSON. The server connects to a PostgreSQL database hosted on AWS RDS.

2 Conventions and Schema Notes

- **Parameter types:** `path` (in URL), `query` (after `?`), `body` (JSON body of POST).
- **SQL bindings:** Listed per route to distinguish URL/body parameters from the values bound into the underlying SQL statement. In most cases the mapping is 1:1; exceptions are noted.
- **review_count vs. recent_review_count:** Two distinct popularity signals.
 - **review_count** — scrape-time all-time review count from the Amazon products dataset (stored on `Products`).
 - **recent_review_count** — count of rows in `Reviews` for an ASIN with `review_timestamp >= NOW() - INTERVAL ' <N> months'`, computed on the fly. Replaces the originally planned `bought_in_last_month` field, which is not present in the US `asaniczka` product dataset.
- **Brand data:** Extracted from product titles during ingestion (no brand column in the source); stored as `Brands` with FK `brand_id` on `Products`.
- **Reviewer stats (total_reviews, avg_rating):** Computed on the fly from `Reviews` rather than precomputed, to preserve a non-trivial runtime for the credibility query (supports the optimization story in Milestone 5).

3 Primary Routes (Query-Backed)

3.1 Product Keyword Search

Retrieves basic product details where the title contains a given keyword. Optionally filters by a minimum star rating. Powers the product search bar on the Home and Browsing pages.

Method / Path: **GET** `/products/search`

SQL Bindings: `:keyword` bound as `'%' || :keyword || '%'` into a LIKE predicate; `:minStars` bound into `Products.stars >= :minStars`.

Request Parameters:

Name	Param Type	Data Type	Required	Description
keyword	query	string	yes	Term matched against <code>Products.title</code> with case-insensitive LIKE.
minStars	query	float	no	Minimum star rating (0.0–5.0). Defaults to 0.

Response Parameters:

Name	Data Type	Description
asin	string	Unique Amazon identifier (PK).
title	string	Product title.
price	float	Current selling price (USD).
stars	float	Average star rating.
is_best_seller	bool	Whether the product has Amazon Best Seller status.

3.2 Category Comparison

Returns per-category aggregate metrics (mean price, mean rating, product count) across the entire catalog. Powers the Category-Wise Comparison analytics view.

Method / Path: **GET** `/analytics/categories/compare`

SQL Bindings: None.

Request Parameters:

None.

Response Parameters:

Name	Data Type	Description
category_name	string	Category display name.
avg_price	float	Mean product price within the category.
avg_rating	float	Mean star rating within the category.
product_count	int	Number of products in the category.

3.3 Shopping Cart Savings

Given a list of ASINs representing the user's cart, returns the total list price, total current price, and total savings. Powers the “Money You Would Have Saved” cart view.

Method / Path: **POST** `/cart/savings`

SQL Bindings: `:asins` bound into `WHERE Products.asin = ANY(:asins)`.

Request Parameters:

Name	Param Type	Data Type	Required	Description
asins	body	array[string]	yes	List of product ASINs in the user's cart.

Response Parameters:

Name	Data Type	Description
total_list_price	float	Sum of original manufacturer prices across cart items.
total_current_price	float	Sum of current selling prices across cart items.
total_savings	float	<code>total_list_price - total_current_price</code> .

3.4 High-Value Alternatives

For a given product, returns up to 5 alternatives in the same category with a higher star rating and lower price than the target. Powers the “Better Alternatives” widget on the product detail page.

Method / Path: **GET** `/products/:asin/alternatives`

SQL Bindings: `:asin` bound into three subqueries to fetch the target's `category_id`, `stars`, and `price`, then used to exclude the target itself.

Request Parameters:

Name	Param Type	Data Type	Required	Description
asin	path	string	yes	Target product's ASIN.

Response Parameters:

Name	Data Type	Description
asin	string	Alternative product's ASIN.
title	string	Alternative product title.
price	float	Current selling price.
stars	float	Average star rating.
category_name	string	Shared category name.

3.5 Brand Performance & Review Sentiment Analytics *(Complex Query)*

Joins `Brands`, `Products`, and `Reviews` to compute brand-level aggregates — distinct best-seller product count, average review rating, and total helpful votes — filtered to brands whose overall product rating exceeds 4.0 and to reviews from verified purchases with at least 10 reviews per brand. Designed for non-trivial runtime pre-optimization.

Method / Path: **GET** /analytics/brands/performance

SQL Bindings: None.

Request Parameters:

None.

Response Parameters:

Name	Data Type	Description
brand_name	string	Brand display name.
total_products	int	Distinct best-seller products for the brand.
avg_review_score	float	Mean rating across qualifying reviews.
total_helpful_votes	int	Sum of helpful votes across qualifying reviews.

3.6 Deals & Discounts Ranking

Returns the products with the largest percentage discount between `list_price` and `price`, capped by a maximum price. Powers the Deals page.

Method / Path: **GET** /deals

SQL Bindings: `:maxPrice` bound into `Products.price <= :maxPrice`.

Request Parameters:

Name	Param Type	Data Type	Required	Description
maxPrice	query	float	no	Maximum current price to include. Defaults to 500.

Response Parameters:

Name	Data Type	Description
asin	string	Product ASIN.
title	string	Product title.
price	float	Current selling price.
list_price	float	Original manufacturer price.
discount_pct	float	$(1 - \text{price}/\text{list_price}) \times 100$, rounded to 1 decimal.
stars	float	Average star rating.
review_count	int	Scrape-time total review count.
category_name	string	Category display name.

3.7 Product Rating Distribution

For a given product, returns the count of reviews at each star level (1–5) along with the verified-purchase ratio per level. Powers the review histogram on the product detail page.

Method / Path: **GET** /products/:asin/rating-distribution

SQL Bindings: :asin bound into `Reviews.asin = :asin`.

Request Parameters:

Name	Param Type	Data Type	Required	Description
asin	path	string	yes	Product ASIN.

Response Parameters:

Name	Data Type	Description
rating	float	Star level (1–5).
review_count	int	Number of reviews at this level.
verified_ratio	float	Fraction of reviews at this level that are verified purchases.

3.8 Most Helpful Reviews with Reviewer Context

Returns the top 10 most helpful reviews for a given product, joined with per-reviewer aggregates (total reviews written, mean rating given) to provide credibility context. Reviewer stats are computed on the fly. Powers the “Top Reviews” section of the product detail page.

Method / Path: **GET** /products/:asin/helpful-reviews

SQL Bindings: :asin bound into `Reviews.asin = :asin`.

Request Parameters:

Name	Param Type	Data Type	Required	Description
asin	path	string	yes	Product ASIN.

Response Parameters:

Name	Data Type	Description
review_id	int	Unique review identifier.
rating	float	Star rating in this review.
review_title	string	Review title.
review_text	string	Review body.
helpful_vote	int	Helpful-vote count.
verified_purchase	bool	Whether the reviewer’s purchase was verified.
review_timestamp	string	ISO 8601 timestamp of the review.
reviewer_total_reviews	int	Total reviews written by this user (computed).
reviewer_avg_rating	float	Mean rating this user has given (computed).

3.9 Top Products by Recent Popularity

Within a given category, returns the products with the highest recent review activity (a proxy for recent purchase volume), alongside their star rating and total review count. Powers the Trending Products view on the Browsing page. Uses `recent_review_count` derived from `Reviews.review_timestamp`.

Method / Path: **GET** `/products/trending`

SQL Bindings: `:category` bound into `Categories.category_name = :category`; `:months` bound into the `INTERVAL` expression on `Reviews.review_timestamp`.

Request Parameters:

Name	Param Type	Data Type	Required	Description
category	query	string	yes	Category name to filter by.
months	query	int	no	Lookback window for recency (default 3).

Response Parameters:

Name	Data Type	Description
asin	string	Product ASIN.
title	string	Product title.
price	float	Current selling price.
stars	float	Average star rating.
review_count	int	Scrape-time total review count.
recent_review_count	int	Reviews in the lookback window (computed).
is_best_seller	bool	Amazon Best Seller flag.
brand_name	string	Product brand.

3.10 Review Trend & Reviewer Credibility Analysis *(Complex Query)*

For a given category, analyzes monthly review trends by joining `Products`, `Reviews`, and on-the-fly reviewer aggregates. Reviews are split into high-credibility (reviewer has ≥ 10 total reviews) and low-credibility groups, then the query returns per-month averages for each group along with the credibility gap. Designed for non-trivial runtime pre-optimization (four-way join, CTE, date truncation, conditional aggregation).

Method / Path: **GET** `/analytics/reviews/trend`

SQL Bindings: `:category` bound into `Categories.category_name = :category`.

Request Parameters:

Name	Param Type	Data Type	Required	Description
category	query	string	yes	Category name to analyze.

Response Parameters:

Name	Data Type	Description
review_month	string	First day of the month (ISO 8601).
total_reviews	int	Reviews posted in the month.
overall_avg_rating	float	Mean rating across all reviewers in the month.
high_cred_avg	float	Mean rating from high-credibility reviewers.
low_cred_avg	float	Mean rating from low-credibility reviewers.
credibility_gap	float	<code>high_cred_avg - low_cred_avg</code> .
verified_count	int	Verified-purchase reviews in the month.

3.11 Top Value Products *(Complex Query)*

Identifies high-value products by comparing each item against its category benchmarks — filtering for products priced below the category average while maintaining a star rating above the category average, with at least one recent review. Powers the Top Value Products view. Designed for non-trivial runtime pre-optimization (correlated subqueries executed per row, plus an existence check on `Reviews`).

Method / Path: **GET** `/products/top-value`

SQL Bindings: `:reviewedSince` bound into `Reviews.review_timestamp >= :reviewedSince` inside the `EXISTS` subquery.

Request Parameters:

Name	Param Type	Data Type	Required	Description
reviewedSince	query	string	no	ISO date; only include products with at least one review on or after this date. Defaults to 12 months ago.

Response Parameters:

Name	Data Type	Description
asin	string	Product ASIN.
title	string	Product title.
price	float	Current selling price.
stars	float	Average star rating.
category_name	string	Category display name.

3.12 Value Score Ranking with Weighted Normalization *(Complex Query)*

Computes a composite value score per product by min-max normalizing four dimensions — star rating, review count, price efficiency (inverse price within category), and recent review activity — and combining them with user-supplied weights. Powers the Value Rankings page with interactive weight sliders. `recent_review_count` replaces the originally planned purchase-volume signal.

Method / Path: **GET** `/products/value-rankings`

SQL Bindings: All four weights bound into the weighted-sum expression in the outer **SELECT**.

Request Parameters:

Name	Param Type	Data Type	Required	Description
wRating	query	float	no	Weight on star-rating dimension (default 0.25).
wReviews	query	float	no	Weight on total-review-count dimension (default 0.25).
wPriceEff	query	float	no	Weight on price-efficiency dimension (default 0.25).
wRecent	query	float	no	Weight on <code>recent_review_count</code> dimension (default 0.25).

Response Parameters:

Name	Data Type	Description
asin	string	Product ASIN.
title	string	Product title.
price	float	Current selling price.
stars	float	Average star rating.
review_count	int	Scrape-time total review count.
recent_review_count	int	Reviews in recent window (computed).
img_url	string	Product image URL.
category_name	string	Category display name.
brand_name	string	Product brand.
value_score	float	Composite weighted score in $[0, 1]$.

4 Auxiliary Routes (UI Support)

These routes support client-side UI (dropdowns, page loads) and are not among the 10 required query-backed routes.

4.1 List Categories

Returns all categories for populating filter dropdowns on the Browsing and Analytics pages.

Method / Path: **GET** /categories

SQL Bindings: None.

Request Parameters:

None.

Response Parameters:

Name	Data Type	Description
category_id	int	Category identifier.
category_name	string	Category display name.

4.2 List Brands

Returns all extracted brands for populating brand filter dropdowns.

Method / Path: **GET** /brands

SQL Bindings: None.

Request Parameters:

None.

Response Parameters:

Name	Data Type	Description
brand_id	int	Brand identifier.
brand_name	string	Brand display name.

4.3 Product Detail

Returns full detail for a single product. Called on initial load of the product detail page before the analytics-specific routes (rating distribution, helpful reviews, alternatives) fire.

Method / Path: **GET** /products/:asin

SQL Bindings: :asin bound into Products.asin = :asin.

Request Parameters:

Name	Param Type	Data Type	Required	Description
asin	path	string	yes	Product ASIN.

Response Parameters:

Name	Data Type	Description
asin	string	Product ASIN.
title	string	Product title.
img_url	string	Product image URL.
product_url	string	Amazon product URL.
price	float	Current selling price.
list_price	float	Original manufacturer price.
stars	float	Average star rating.
review_count	int	Scrape-time total review count.
is_best_seller	bool	Best-seller flag.
category_name	string	Category display name.
brand_name	string	Brand display name.

5 Summary

#	Route	Method	Query-Backed	Complex
1	/products/search	GET	✓	
2	/analytics/categories/compare	GET	✓	
3	/cart/savings	POST	✓	
4	/products/:asin/alternatives	GET	✓	
5	/analytics/brands/performance	GET	✓	✓
6	/deals	GET	✓	
7	/products/:asin/rating-distribution	GET	✓	
8	/products/:asin/helpful-reviews	GET	✓	
9	/products/trending	GET	✓	
10	/analytics/reviews/trend	GET	✓	✓
11	/products/top-value	GET	✓	✓
12	/products/value-rankings	GET	✓	✓
13	/categories	GET	auxiliary	
14	/brands	GET	auxiliary	
15	/products/:asin	GET	auxiliary	

Total: 12 query-backed routes (exceeds the 10 minimum), 4 marked complex, 3 auxiliary routes.